INCH-POUND

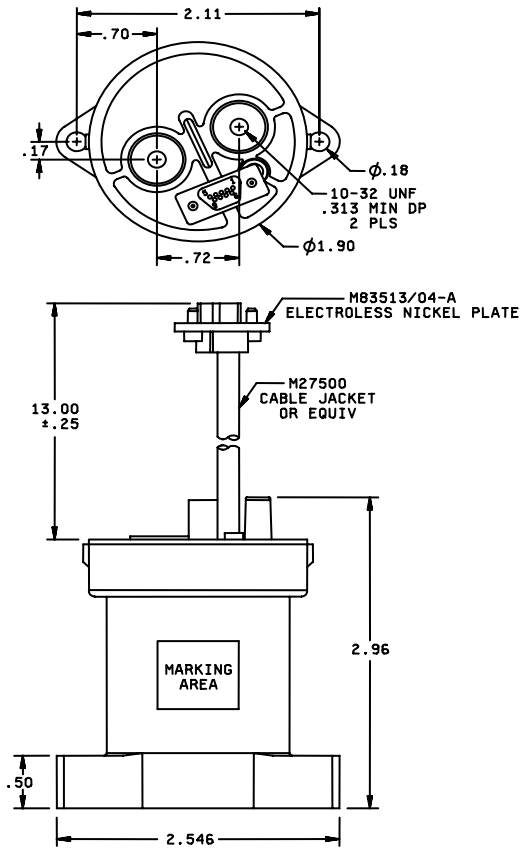
MIL-PRF-32085/2
11 May 2006

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, SPST, 270 VOLT DC,
100 AMPERE, HERMETICALLY SEALED,
ELECTRICALLY HELD OR MAGNETIC LATCHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-32085.



Inches	mm	Inches	mm
.020	0.51	.800	20.32
.060	1.52	1.05	26.7
.10	2.5	1.125	28.58
.15	3.8	1.380	35.05
.150	3.81	1.810	45.97
.215	5.46	1.927	48.95
.221	5.61	2.25	57.2
.250	6.35	2.36	59.9
.37	9.4	3.37	85.6
.40	10.2	3.48	88.4
.502	12.75	3.88	98.6
.566	14.38	12.0	305
.615	15.62		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on all dimensions is ±.030 inch (0.77 mm).

FIGURE 1. Dimensions and configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:38Z
Check the source to verify that this is the current version before use.

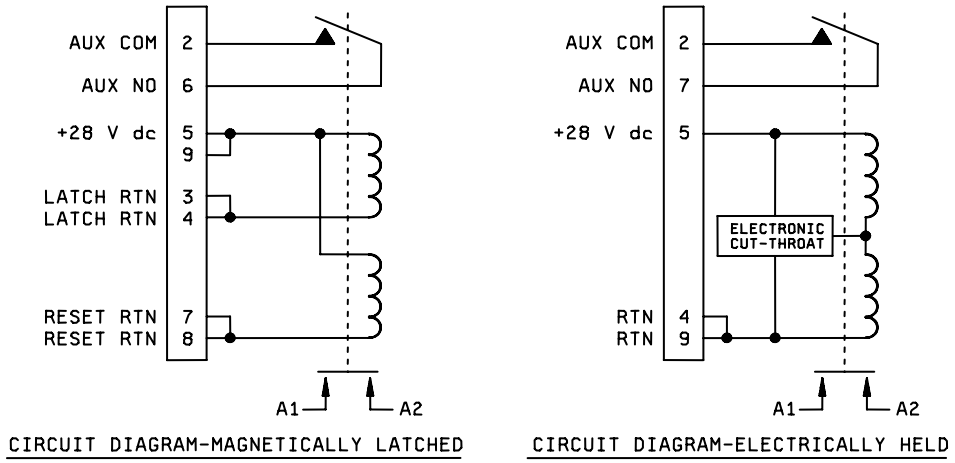


FIGURE 2. Circuit diagrams.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Terminations: See figure 1.

Configuration: 1PST: Magnetic latch or electrically held.

Weight: 0.8 pound maximum.

Seal: Hermetic

Finish: Flat black or off-white.

Rated coil voltage: 28 V dc.

Maximum coil voltage: 32 V dc.

Pickup voltage (magnetic latching and electrically held):

Nominal: 16 V dc at 25°C.

Nominal: 18 V dc at -40°C to +71°C.

High temperature test: 19 V dc.

Continuous current test: 22 V dc.

Reset voltage (magnetically latching):

Nominal: 16 V dc at 25°C.

Nominal: 18 V dc at -40°C to +71°C.

High temperature test: 19 V dc.

Continuous current test: 22 V dc.

Dropout voltage (electrically held): 1.5 V dc minimum.

Hold voltage (electrically held): 8 V dc minimum, -40°C to +71°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:38Z
Check the source to verify that this is the current version before use.

Voltage rating:

- Main contacts: 270 V dc.
- Auxiliary contacts: 28 V dc.

Current rating:

- Main contacts:
  - Resistive: 100 amperes.
  - Inductive: 75 amperes.
  - Motor: 50 amperes.
- Auxiliary contacts:
  - Resistive: 0.1 amperes at 28 V dc.

Coil resistance at 25°C, ±10 percent.

- Electrically held:
  - Pickup: 13 ohms.
  - Pickup + hold: 118 ohms.

Magnetic latch:

- Latch: 18 ohms.
- Reset: 14 ohms.
- Contact voltage drop:
  - Electrically held:
    - Before life: 175 mV maximum.
    - End of life: 250 mV maximum.
  - Magnetic latch:
    - Before life: 100 mV maximum.
    - End of life: 150 mV maximum.

Operate time (latch and electrically held): 40 ms maximum.

Reset time (magnetic latch): 15 ms maximum.

Release time (electrically held): 15 ms maximum.

Contact bounce time (main and auxiliary): 5 ms maximum.

Duty rating:

- Main and auxiliary contacts: Continuous.
- Coil (electrically held): Continuous.
- Coil (magnetic latching): 20 percent.

Overload current: 500 amperes dc.

Rupture current: 1,500 amperes dc, 25 cycles.

Coil transient suppression (electrically held): 42 V dc maximum.

Life: Resistive, motor, and inductive, 25,000 cycles each.

Mechanical life: 100,000 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:38Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage:

Sea level: 1,500 V rms from main contacts (closed) to all other points.

70,000 feet: 500 V rms from main contacts (closed) to all other points.

Temperature range:

Operating: -40°C to +71°C.

Non-operating: -55°C to +85°C.

Vibration:

Sinusoidal: 15g in accordance with method 204 of MIL-STD-202.

Frequency range: 10 to 2,000 Hz.

Random: 8.5g rms overall, in accordance with method 214 of MIL-STD-202.

Frequency (Hz)	PSD(g^2/Hz)
10	.01
10-50	+3 dB/Oct
50-1,000	.05
1,000-2,000	-7 dB/Oct

Acceleration: 15g.

Maximum altitude rating (operating): 70,000 feet.

Shock: 20g's in accordance with method 213 of MIL-STD-202.

Marking: Marking shall be in accordance with MIL-PRF-32085.

Part or Identifying Number (PIN): M32085/2- (dash number from table I and suffix letter designating failure rate level.)

TABLE I. Dash numbers and type of loads tested during life.

Dash number	Type of loads tested during life
Magnetically latched	
100	Resistive
101	Resistive and motor
110	Resistive and inductive
111	Resistive, inductive, and motor
Electrically held	
200	Resistive
202	Resistive and motor
220	Resistive and inductive
222	Resistive, inductive, and motor

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:38Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-PRF-32085 this document references the following:

MIL-STD-202 MIL-DTL-83513/4 NEMA-WC-27500

Custodians:
Army - CR
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2006-008)

Review activities:
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T04:38Z
Check the source to verify that this is the current version before use.